

CrowdAnonymizer



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Programm Introduction

This application provides tools to record anonymized videos or to anonymize existing videos and images. It includes real-time human-head detection using a custom YOLO model that identifies heads from any direction and applies a blur effect. For stronger anonymization, all recorded videos are converted to grayscale. The program supports one or two webcams to achieve a wider field of view. Videos are saved in **.avi** format, and only fully anonymized frames are saved.

Use as python script

To run the application as a Python script, you need to install the following packages:

- opencv-python
- numpy
- ultralytics
- tkinter

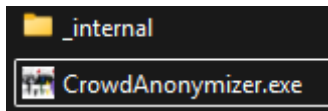
Due to compatibility issues with Python **3.14**, it is recommended to use **Python 3.12.7**. Use the following library versions for best stability:

numpy 2.0.1, opencv-python 4.9.0.80, ultralytics 8.1.0

Place the background image, the application icon, and the YOLO model file human_head.pt in the same folder as main.py.

Use as .exe

When using the compiled .exe version, you must place the folder _internal in the same directory as the .exe file.



The _internal folder contains all required files, including:

- the background image
- the application icon
- the YOLO model human_head.pt
- all bundled libraries

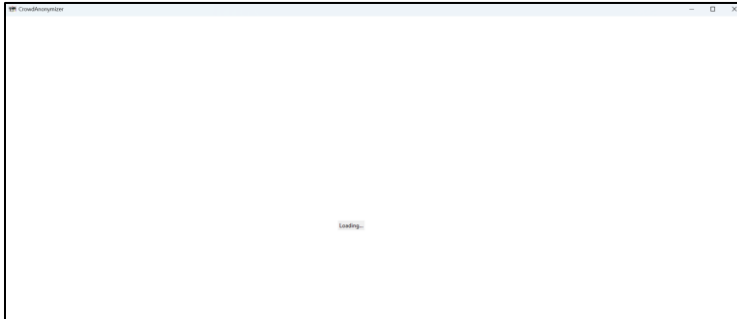
Currently, the folder is very large because the entire libraries are included. This results in longer loading times.

Important note: Due to a bug, after starting a video recording, a new window labeled “loading” may appear. You must close this window manually. After that, the program will function normally.

Programm Execution

1. Loading Screen

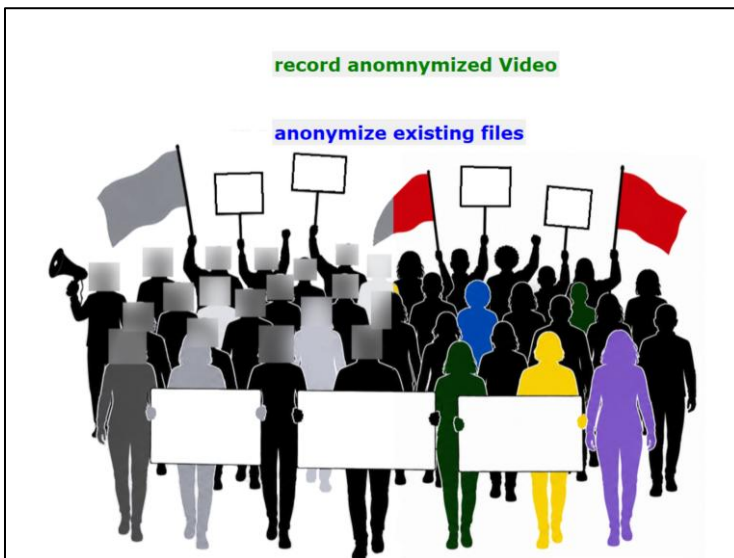
The application starts with a loading window.



2. Select Mode

Click on the text to choose between:

- **Recording anonymized video**
- **Anonymizing an existing file**



Anonymizing existing File

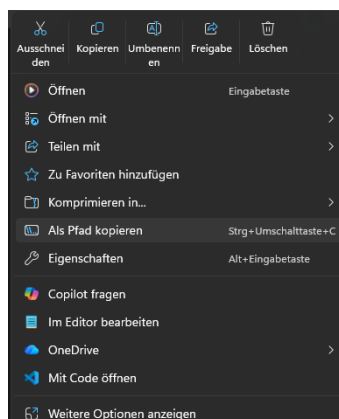
You can paste the path to a video or an image. The program will load the file, anonymize it, and save the processed version in: user/Pictures/CrowdAnonymizer

You can choose:

- color mode (grayscale or color)
- output format



Right-click the file and select "Copy as link".



Anonymize Video

- Choose the output format (.avi or .mp4)
- Choose whether the video should be converted to grayscale or remain in color

Anonymize Image

- Choose whether the image should be converted to grayscale or remain in color

Start the process by clicking the green play button. After processing is complete, the play button will reappear.



Record anonymized Videos



- Select one or two cameras
- If two cameras are selected, their frames will be combined side-by-side into one video
- Choose the camera numbers assigned by your system
 - -1 means the camera is not used

Recorded videos are automatically named:

out-DDMMYYhhmm.avi (*DDMMYYhhmm* = *day, month, year, hour, minute*)

All recorded videos are saved:

- in **grayscale**
- in **.avi format**

(You can convert them to .mp4 later using the “anonymize file” option.)

Start recording by pressing the green play button. During recording, the button turns into a red circle. Click the red circle to stop recording.

A timestamp is added in the lower-right corner of the video.

Videos are saved in: user/Pictures/CrowdAnonymizer

